

Project Initialization and Planning Phase

Date	2 July 2026
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to develop an intelligent flood prediction system that uses Machine Learning algorithms to accurately predict flood risk based on environmental conditions. The system enables early warning and supports disaster preparedness.
Scope	he project focuses on collecting environmental data, training a Machine Learning model, predicting flood occurrence, visualizing flood risk through graphs, and providing predictions using a simple Flask-based web application
Problem Statement	
Description	Floods are one of the most common natural disasters, causing significant damage to human life, agriculture, infrastructure, and the environment. Existing warning systems often lack timely and accurate predictions.
Impact	An effective flood prediction system can: • Provide early flood warnings. • Reduce loss of life and property.
Proposed Solution	
Approach	The proposed system uses Machine Learning techniques to analyze historical environmental data, including rainfall, water level, humidity, and temperature. A K-Nearest Neighbors (KNN) model predicts whether a flood is likely, while K-Means Clustering groups similar flood-risk conditions

Key Features	Machine Learning-based flood prediction using KNN. Flood risk clustering using K-Means. Visualization of rainfall and water level using graphs
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	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv